🧾 🚀 HiveQL Cheat Sheet (Flight System)

🟢 1. Start Hive

hive

Exit:

exit;

🟡 2. Database Commands

```
CREATE DATABASE flight_db;
SHOW DATABASES;
USE flight_db;
DROP DATABASE flight_db;
```

🔵 3. Create Table (Internal)

```
CREATE TABLE flights (
  year INT,
  month INT,
  day INT,
  flight_no STRING,
  origin STRING,
  destination STRING,
  dep_delay INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

🟣 4. External Table

```
CREATE EXTERNAL TABLE flights_ext (
  year INT,
  month INT,
  day INT,
  flight_no STRING,
  origin STRING,
  destination STRING,
```

```
  dep_delay INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
LOCATION '/user/cloudera/flight_data';
```

📤 5. Load Data

```
LOAD DATA INPATH '/user/cloudera/flight_data/flights.csv'
INTO TABLE flights;
```

➕ 6. Insert Data

```
INSERT INTO flights VALUES (2008,1,4,'F107','NYC','LAX',25);
```

🧱 7. Alter Table

```
ALTER TABLE flights ADD COLUMNS (airline STRING);
```

🔍 8. View Table

```
SELECT * FROM flights;
DESCRIBE flights;
```

🔗 9. Join Tables

```
SELECT f.flight_no, a.airline
FROM flights f
JOIN airlines a
ON f.flight_no = a.flight_no;
```

📌 10. Create Index

```
CREATE INDEX flight_index
ON TABLE flights (flight_no)
AS 'COMPACT'
WITH DEFERRED REBUILD;
```

Rebuild:

ALTER INDEX flight_index ON flights REBUILD;

📊 11. Aggregate Query (IMPORTANT)

SELECT day, AVG(dep_delay) AS avg_delay
FROM flights
WHERE year = 2008
GROUP BY day;

🧹 12. Drop Table

DROP TABLE flights;

🧠 🔥 Important Concepts (Viva)

👉 Internal vs External Table

- Internal → data deleted on drop
- External → data remains

👉 HiveQL

- SQL-like language for Hadoop

👉 Index

- Improves query performance
- Rarely used in real-world Hive

👉 Load vs Insert

- LOAD → from HDFS
- INSERT → manual values

⚡ Common Errors + Fix

Error	Fix
Table not found	USE flight_db;
Output empty	Check data load
Index error	Check column name
Hive error	Start metastore

🎯 One-Line Summary (Exam Gold)

“HiveQL is used to perform SQL-like operations on large datasets stored in Hadoop.”